

Simeulue, Nias and Siberut, Indonesia

Legend, Ritual and Architecture on the Ring of Fire

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Abstract

Indigenous practices have proven to reduce the impacts of natural disasters in three islands in Sumatra, Indonesia, namely Simeulue, Nias and Siberut. Though culturally different, all three islands have within the last five years experienced earthquakes and tsunamis and have brought to light indigenous practices previously unremarked by the international community concerned with disaster risk reduction. These practices include traditional communication, human-settlement planning and building methods and associated rituals. These practices will be examined in detail to fully understand their impact and consider what relevance these indigenous practices and knowledge have for modern development.

Background

Simeulue

Part of the Province of Aceh, Simeulue is an island district which consists of 1 large island, i.e. Simeuleu Island, and approximately 40 small islands. The islands cover a total area of 205,148.63 ha and are located approximately 155 km from the mainland Sumatra.

Nias

Nias consists of a group of islands located between Simeulue and the Mentawai islands, approximately 100 kilometers off the west coast of North Sumatra, Indonesia. The islands cover an area of 4,771 km². Administratively, Nias Island group belongs to the North Sumatra Province and is divided into two districts, Nias and Southern Nias. According to the population census of 2006, the total population of Nias

is estimated at 713,045¹. A view of a village in Nias is shown in Figure 1.

Siberut

With an area of 400,030 ha, Siberut is the largest of the Mentawai islands which encompasses more than seventy islands and islets. It is located off the west coast of the province of West Sumatra. The more than 35,000 indigenous inhabitants of Siberut, belonging to the Mentawai ethnic group, represent one of the few remaining communities in Southeast Asia whose way of life is still to a large extent dependent on the natural environment. Many parts of Siberut still feature a semi- subsistence economy in which indigenous environmental management and related belief systems play an important role in the daily lives of peoples. A typical Siberut village is displayed in Figure 2.

¹ Badan Pusat Statistik Sumatera Utara

Story/Event

Within the last five years, Simeulue, Nias and Siberut have borne the brunt of several earthquakes and tsunamis. The December 2004 Tsunami affected both Simeulue and Nias. Simeulue, however, only suffered a small number of casualties compared to other areas. The official report issued by the district government declared only seven casualties out of a total population of over 78,000, nearly 95% of whom live in coastal areas. When the earthquake occurred on 26 December 2004, the inhabitants of Simeulue knew that they had to evacuate to higher ground, as there was the possibility of a tsunami. This reaction to the earthquake minimized the potential devastating impact of the tsunami. Besides the indigenous knowledge, the hilly topography of the island was also another important factor that contributed to a small number of casualties. The hills are located only a few hundred meters from the villages and the shoreline.

Nias Island was strongly affected by the 26 December 2004 earthquake and by the following tsunami that killed 140 islanders and rendered hundreds homeless³. The earthquake that hit a few months later, on 28 March 2005, measured a devastating 8.7 on the Richter scale and claimed 839⁴ lives. The impact of the earthquake was so enormous that some areas of the island were uplifted by more than 2 meters, exposing coral reefs and other areas as much as 100 meters seawards. The earthquake affected the lives of 90% of the population, leaving 15,000 houses needing rehabilitation and 29,000 needing reconstruction⁵.

² UNORC (2005).

³ Badan Rekonstruksi dan Rehabilitasi (BRR) NAD – Nias, (2005, October)

⁴ BRR (2005, October).

⁵ BRR (2005, October).



Figure 1. A village in Nias.



Figure 2. A Siberut village.

On 12 September 2007, an earthquake measuring 7.9 on the Richter scale occurred near Siberut. On the entire island there was only one casualty. One of the reasons for the low death toll was that most people left their houses as soon as they felt the earthquake and took refuge in an open space. Such collective reaction to earthquakes can be attributed to the public knowledge communicated through legends.

Indigenous Knowledge

Simeulue

The low death toll on Simeulue Island can be attributed to the community's indigenous knowledge. According to local people, this knowledge can be traced back to an "ancestral experience" in 1907, when an earthquake generated a tsunami and killed a large number of the island's dwellers. Stories from the disastrous 1907 event were translated into stories, testimonial monuments and reminiscence, which were then transferred to younger generations in an unstructured and ad-hoc pattern.

The Simeulue community uses the word *smong* to describe this event, which has the same meaning as tsunami. The existence of a local name to depict a tsunami event shows that the local community possesses a certain level of knowledge in regard to the phenomenon. Though it is unclear whether the word *smong* existed before the tsunami event of 1907, many people on Simeulue believe strongly that it did and that it is derived from *ni semongan* or splashing (of water) in one of the three local languages. The current

interpretation of smong by the local islanders is that it consists of a linear series of natural events that start with strong earthquakes, a receding sea and then followed by giant waves and flooding.

The events occurring during the 1907 tsunami were transferred orally through stories within the community without an outlined structure. The story telling was not aimed at preparing new generations for future similar events, but instead gave a description of a historical event. The event was so deeply traumatic that it was translated into a series of individual stories, which were transferred at both the family and community level. Each story covered the event in its own way and often referred to the suffering or death of family members. Since the 1907 tsunami happened on a Friday, at the time the men were coming back from the mosques after their Friday prayers, the event took on a symbolic significance which led the Islamic community of Simeulue to also perceive the 1907 event as the wrath of God. The event is often referred to as a means by God to rectify behavior of family members which depraved the existing social and religious norms. The stories also refer to natural tsunami monuments such as the presence of coral in the inland rice fields.

Nias

The earthquake of 28 March 2005 had a huge impact on Nias island's inhabitants and infrastructure. Amidst this devastation a solid example emerged of how indigenous practice can reduce the impact of disaster. After the earthquake, most of the few remaining traditional wood houses had resisted the impact of the earthquake.

As shown in Figure 3, the traditional houses in Nias are built of wood, where the structural members are slotted together rather than nailed. This technique adds to the flexibility of the house. The houses are supported by vertical pillars (enomo), which rest on stone blocks and a number of diagonal beams (ndriwa). The ndriwa are placed between the enomo both across and along the building, functioning as longitudinal and lateral bracing. They lean against each other at their base and top where they fit into the horizontal beams located under the floor of the building. The technique of

diagonally bracing the ndriwa in both directions allows the structure to withstand the strongest of tremors.

The roof of the building has a similar construction as the understructure of the house. Depending on the size of the building the roof will have several layers of diagonal beams which are placed in a lateral position and lean against vertical and horizontal beams at both ends. Both the understructure and the roof provide a three-dimensional structure that enhances the elasticity and stability of the house when an earthquake occurs. This feat of engineering is still considered to be the most effective example of a tried and tested earthquake resistant architecture on Nias Island.

Siberut

The Mentawai community's past individual and collective experiences with natural phenomena like earthquakes have been translated into tangible and intangible strategies that reduce the impacts of earthquakes. For example, one of the intangible strategies is a legend highlighting the metaphysical relationship between peoples and earthquakes. The legend provides a detailed description of the first earthquake ever to occur. Besides raising the awareness regarding the dangers of the earthquake, it also reminds the listener of the human origin and emotions of the earthquake.

The following is a version of the legend as heard in Attabai, a small settlement in southern Siberut:



Figure 3. The traditional house in Nias, a time-tested and effective earthquake resistant structure.



Figure 4. Shamen are the instruments by which traditional belief is passed on.

“Once upon a time in one of the valleys in the west coast of Siberut Island lived an extended family that planned to build a big new ‘longhouse’ (uma). Preparations and efforts were made to build the uma: large trees were cut down for the pillars, sago-palm leaves were weaved for roofing, timber was cut for the flooring, and rattan vines were made into ropes. To ensure that there was enough manpower to construct the uma, the extended family invited many relatives and neighbours living in other valleys to help.

After all the preparatory work was finalized, it was time to dig the hole for the first pillar. After the whole was dug, the owner of the uma dropped his chisel into the hole and asked his brother-in-law to pick it up. The owner of the uma did not like his brother in law and when his brother-in-law was in the hole, the owner ordered other members of his extended family to entrench the huge pillar into the hole. A faint moan of tremendous pain was heard and after a strong thrust on the huge pillar, only silence remained.

The construction of the uma was finalized and was renowned for its beauty all over the island. The time had come for inaugurating the house through a big ceremony and festivity, and the owner of the uma invited many relatives and neighbours to participate in the feast. Many pigs and chickens were slaughtered and cooked for a large meal.

The spirit of the brother-in-law who could not rest in peace was very angry at the owner and the people participating in the feast. Before the celebration, he came to warn his sister and her children that during the party, they should not eat in the uma and instead hide under a banana tree.

When everyone started eating, the ground suddenly shook extremely strong, and a thundering noise came from below the ground. The earthquake was the vengeance of the spirit. The newly built uma collapsed and all people were killed, except from the family of the spirit. They were the only survivors.

From that day on, according to the story teller, the Mentawaiers named the earthquake “Grandfather” (teteu).”

There are actually two words for earthquake in Mentawaiian language, *sigegeugeu* and *teteu*. *Sigegeugeu* can be translated as “a shaking” and has very much the same connotation as the word earthquake in English. The other name ‘*teteu*’, or grandfather has clearly a metaphysical meaning. It reflects an anthropomorphization of a possibly lethal natural phenomenon that needs to be approached with a certain level of respect and fear. According to the traditional belief system, each time a new house is built, people must give offerings to the spirits of the earth (*Taikabaga*) to avoid the *teteu* (grandfather) from becoming upset and angry. The offerings are put near the important pillars of the house to appease the earth spirits and prevent the house from collapsing if *teteu* strikes again. The act of making the offering serves to subconsciously reinforce the community’s collective awareness of the risk of earthquakes and keeps it alive in the consciousness. Figure 4 illustrates the shamen at a ceremony.

Lessons Learned

Simeulue

It is important to highlight some key characteristics of the traditional story telling in Simeulue that saved so many lives in 2004. Firstly, the stories are concerned with local information. Secondly, the story-telling takes place in the home from one family member to another. These two factors together serve as an effective communication method to inform a community about the risk of disaster in their locality. Modern communication methods are less focused on information pertaining to the local area and disseminate information in a much broader manner. With the introduction of modern communication channels, such as radio, television and telephone comes a broadening of horizons and a wider scope of information. Lessons are taken from the world beyond instead of from the past or the specific local context. In this instance, new means to inform communities about the threats of disaster in their localities must be found. These means must carry relevant information and must also replicate the effectiveness of the oral story-telling tradition in dissemination and immediate relevance to local communities. In Simeulue, a bedtime story about an event in local history saved hundreds of lives. Modern development programs, which seek to prepare communities through brochures and focus-group discussions, may yet have something to learn from indigenous story telling traditions and practices.

Also it is important to remember the part played by the topography of the island of Simeulue. Villagers, armed with their advanced knowledge of the threat of a tsunami, were able to access hills to where they escaped to safety. This indicates the importance of the role of human settlement planning in disaster risk mitigation. Had there been no access to safe areas, knowledge of the legend in 1907 would not have saved so many. This is a key lesson for future disaster risk reduction efforts – a holistic approach needs to be taken towards protecting communities from disasters. Furthermore, risk reduction strategies must be integrated in all aspects of the community's existence.

Nias

Since the March 2005 earthquake, the indigenous practices used to construct traditional houses in Nias have been studied and have gained a reputation for their earthquake resistant quality. However, whilst this example of indigenous practice is celebrated, the devastating impact of the March 2005 earthquake on Nias Island, its population and its economy, must not be forgotten. This seems ironic given that Nias Island was at the time and still is home to one of the world's best example of earthquake resistant architecture. This irony shows us firstly that isolated examples of indigenous practice alone cannot contribute significantly to disaster risk reduction. Secondly, it demonstrates that as the traditional is given up in favor of the modern, communities can be left exposed to the risk of disasters. Modernization has played a big part in the rapidly disappearing traditional architecture of Nias. The status symbol which is represented by this shift to modern design and lifestyle is adequate reason for most people to choose the less resistant Malayan houses over the traditional wooden structures. Deforestation has exacerbated the situation. The hardwood needed to build traditional houses is in scarce supply. As a consequence many of the methods and techniques used to build traditional houses are slowly being forgotten since concrete and bricks have replaced timber as construction material.

Siberut

The story of teteu is part of the traditional belief system of the Mentawai people. The local communities make frequent reference to traditional legends and songs which had shaped their reactions to the earthquake when it was felt. In many cases, people were saved because of the knowledge they had gained from listening to these legends and songs.

The legend which helped reduce the casualties resulting from the earthquake may not be considered a disaster risk reduction method in itself, since its impact is inextricably linked to traditional house building rituals;

however, the traditional belief system and related legends do contribute to disaster risk reduction. With the pressure of development, these may lose significance and impact within the local communities. The adoption of monotheistic religions has started to erode people's belief in spirits, including teteu, and the ritual offerings made to such spirits are not as frequent as before. It was seen how the act of making offerings serves to reinforce knowledge of earthquakes in the collective memory of the communities. Yet as these practices are replaced by the introduction of more modern beliefs, this knowledge is on the brink of being forgotten. Therefore, in order for this knowledge not to be lost, there is a need for modern alternatives which emulate the same transfer of knowledge with similar significance to the local communities.

The three examples of indigenous practices in disaster risk reduction illustrated here have one important common thread: they are all found in remote, underdeveloped areas, which is often but not always the case with indigenous practice. In the cases of Simeulue, Nias and Siberut these examples still exist in practice today precisely because the islands have been cut off from the more developed main land and they have largely been exempt from modern development. They have all been minimally touched by modern infrastructure, education, communication and transportation methods. Development has only just begun to make inroads into the three islands, but already the effects can be seen on indigenous disaster risk reduction practices and knowledge. The question that poses itself here is how will development and modernization affect indigenous practice as progress continues? With regards specifically to disaster risk reduction, what is role of indigenous practices in the modernization of these three islands? Should efforts be made to preserve these isolated examples of indigenous practice or is it naïve to think of these practices as easily packaged solutions that can be reproduced for the benefit of future generations?

The pace and nature of modern development does not always leave room for indigenous practices. When indigenous practice is not replaced by a modern alternative, communities can become more vulnerable to disasters. Indigenous practices have allowed many communities over the centuries to live alongside disaster, but when modernization progresses too rapidly the value of these indigenous practices within this new, modern context is not always understood. It is only when development progress is interrupted by nature in the form of devastating disasters that examples of indigenous practices are once again thrown into the spotlight and it will be remembered

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Dagupan City, Pangasinan, Philippines

Combining Indigenous and Scientific Knowledge in the Dagupan City Flood Warning System

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Abstract

The kanungkong is a bamboo instrument which was traditionally used to call community members to assemble at the village hall for meetings, alert people or call children home. The flood early warning system set up in eight villages in Dagupan City, Philippines, has revived the use of the kanungkong along with staff gauges as flood markers in strategic locations in the villages of the city. The indigenous knowledge is combined with modern scientific knowledge and equipment for use in disaster risk reduction.

Background

Eight flood-prone barangays (villages) in Dagupan City in the province of Pangasinan in northwestern Philippines, namely Mangin, Salisay, Tebeng, Bacayao Norte, Bacayao Sur, Lasip Grande, Lasip Chico and Pogo Grande, have incorporated the use of the kanungkong to relay warning messages to households in the communities, especially those living along the riverbanks. These villages prioritized flood preparedness and mitigation activities under the Program for Hydro-meteorological Disaster Mitigation in Secondary Cities in Asia (PROMISE) project. The community members held workshops to discuss the early warning system and had community drills.

Story/Event

Dagupan City is prone to heavy flooding. In 2007, typhoons with monsoon rains hit Northern and Central Luzon in August and November causing swelling of the river system in Dagupan City. This event put the kanungkong based early warning system to the test. Because the Barangay Disaster Coordinating Council (BDCC) monitored the flood markers and reported these to the Emergency Operations Center of the City Disaster Coordinating Council (CDCC), the villages were prepared in case heavy flooding occurred. The system successfully allowed the entire community to prepare and respond to the disaster with sufficient time.



Figure 1a & b. The kanungkong is a communication medium in the village early warning system.

Indigenous Knowledge

The kanungkong is a communication device which was used extensively in the past by the people of Dagupan City, nearby municipalities and provinces in Northern Luzon (Figures 1a & b). Its uses include calling the community to assemble at the village hall for a meeting, alerting citizens of a robbery incidence during the night, calling attention to midwives to assist a pregnant woman getting ready to give birth, and calling children to come home. With modern ways of communication, it has been forgotten.

Kanungkong comes from the word mangkanungkong which literally means to make a sound. The kanungkong is made of bamboo and when hit with a stick produces the sound kung, kung, kung. The village-level early warning system uses the kanungkong as a local relay and communication medium. For flood monitoring, and as the basis for the relaying of messages, staff gauges or flood markers have been put up and are monitored in strategic locations in the villages.

Presently, people have become familiar with the warning codes adopted in the city, which conform to international disaster warning color standards. In order to put the kanungkong into a system, an agreed upon rhythm and sound (i.e. number of strikes of the kanungkong at designated time intervals) were made to correspond to specific actions. One kanungkong for every 5 houses relays the warning to the households along the river banks. Table 1 explains the warning codes.

Staff gauges have been constructed at the lowest point in the barangays to correspond to the alert warning, based on information from past flooding in the communities (Figures 2a, b & c). A common zero point was initially suggested to be standardized by the city government but the system now operates with each barangay having

Table 1. Warning codes adopted in Dagupan City

Color	Alert Level	Warning Signal by the Kanungkong
White (Ready)	Normal	
Yellow (Get Set)	Alert (warning, there is danger)	5 strikes of the kanungkong at 20 minutes interval
Orange (Go)	Prepare for evacuation or proceed to holding area (heavy flooding approaches)	10 strikes at 20 minutes interval
Red	Full evacuation (evacuate from homes to designated safe areas) Forced evacuation	Non-stop (15 strikes at 10 minutes interval) Non-stop (20 strikes at 5 minutes interval)
Green	Back to normal	

its own agreed upon flood markings at strategic places in the barangay. The gauges show critical levels to determine when people have to prepare to move out and proceed to evacuation centers.

The monitoring and relay of information of flood level from the staff gauge is done by the barangay warning and communications team by hand-held radio to the BDCC. The kanungkong is then sounded and relayed from point to point (every 5 houses apart) (Figure 3). Each BDCC has a radio connection with the CDCC, and information is relayed from one to the other through its Emergency Operations Center by radio.

The flow of the early warning system is illustrated in Figure 4.

The Emergency Response and/or Disaster Risk Management Plan details the responsibilities of the CDCC and BDCC committees and personnel with regards to warning and evacuation. As part of the plan, the early warning system has been drawn up through a series of consultations, study visits and workshops.



Figure 3. The Barangay Disaster Coordinating Council gives out the initial warning using the kanungkung



Figure 2a, 2b & 2c. Staff Gauges showing warning levels monitored by warning and communications team

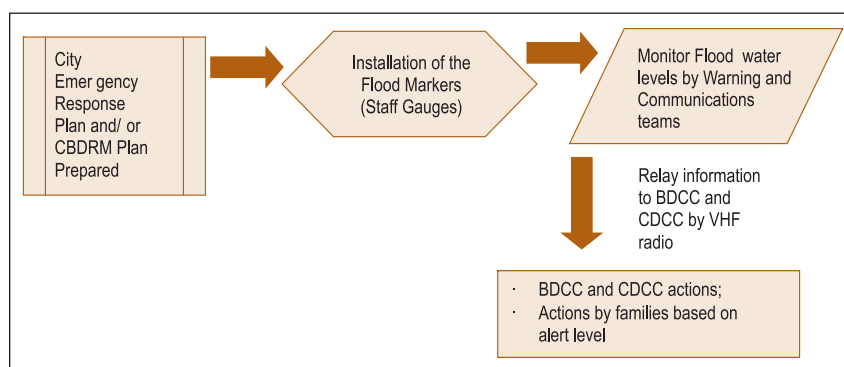


Figure 4. Relay of information to the community through the kanungkung.

Lessons Learned

The Dagupan City flood warning system, a combination of indigenous and modern scientific knowledge, is an effective response to the perennial problem of flooding in the city. In formulating the system, some important lessons were realized which are as follows:

1. The use of the kanungkong has mobilized local capacity while reviving and maintaining a local practice which is now used in disaster preparedness.
2. It is important to involve the community in risk assessment (i.e. hazard, vulnerability and capacity assessments) and designing of the early warning system.
3. It is important to test the warning system and evacuation procedure through table top exercises and practical community drills.
4. “Learning from good practice” visits to the communities involved in community based disaster preparedness and mitigation encourages the communities as well as local government officials to continue the good work. Study visits by local officials and community members to similar projects afford critical reflection on how to improve their own preparedness and mitigation activities.

Barangay Matanag, Legazpi City, Albay, Philippines

Indigenous Know-How on Mayon Volcano's Lava-Spittle Mysticism

Gerardine Cerdena

Abstract

Residing near a volcano does not mean people cannot thrive on its surroundings. In the case of Barangay (village) Matanag, inhabitants stand defiant against volcano disasters and still call the volcano home. The local knowledge acquired within this community about warning signs and how to predict volcanic eruptions helps to minimize risk and cope with the dangers of the Mayon volcano.

“If Mayon is spitting fiery sparks, that is the signal that it will erupt. Sometimes we worry about it, sometimes not. The volcano always makes low rumbling sounds and the farmers are the ones who hear it first.”

Domingo Arias, Barangay Matanag resident

Background

One of the active volcanoes on the Philippine island of Luzon, Mayon is considered by some as the world's most perfectly formed volcano because of its symmetrical cone (Figure 1). It is one of the volcanoes in the “Pacific Ring of Fire” located around the lip of the Pacific Ocean where volcanic activity and earthquakes are common. The volcano is situated 15 kilometers northwest of Legazpi City, Albay, Philippines in the Bicol Region.

Mayon Volcano is a basaltic-andesitic volcano located between the Eurasian and the Philippine plate that was developed through alternate pyroclastic and lava flows spouting out ashes in the past 400 years (Figure 2).

The upper slopes of the volcano are precipitous and rugged, averaging 35-40 degrees, and are capped with a small crater. Its sides are layers of lava and other volcanic material. Magma is formed where the rock melts and eruption usually begins with lava flowing like a geyser from a long crack in the cone. Barangay (Village) Matanag in Legazpi City, a farming village which comprises of 1,400 residents, is one of the lava-prone areas flanking a slope of Mayon Volcano. It was declared a danger zone by volcanologists from the Philippine Institute of Volcanology and Seismology through the use of remote sensing and satellite relays.

Story/Event

Mayon has had forty-seven eruptions in recorded history. The most fatal eruption of Mayon occurred on 1 February 1814 wherein 2,200 people were killed and volcanic ash buried the town of Cagsawa. In Mayon's last fatal eruption in 1993, pyroclastic flows killed 77 people, mainly farmers.

Mayon eruptions often trigger people's fear of losing their homes and crops. On the impact of Mayon's eruption, Bienvenido Belga Sr., Barangay Captain of Matanag shares, "An niyog tapos mga gulayon nagkakaaralang tapos nagkakagaradan pag nagtutuga an Mayon. Su mga gapo nagdadalagasan nin mga kalayo hali sa Mayon kaya minsan nagigipit kami sa negosyo pagmay eruption. – Coconuts and vegetables wilted and died due to Mayon's eruption. Hot rocks fall heavily from Mayon so our businesses were affected." The town solely depends on the locals' coconut and rice products for income and livelihood. In addition, mudslides of volcanic ash and boulders off of Mayon can kill hundreds and can cover a large portion in mud up to the village houses' roofs. In contrast, eruptions can also be viewed as favorable since many inhabitants know that eruptions of volcanic ash also enrich the soil, producing better crops. In fact, threats of volcanic activity and eruption over the years have not dampened the Matanag villagers' spirits.



Figure 1. Mayon Volcano with its perfectly symmetrical cone.
Photo: Jenny Exconde



Figure 2. Pyroclastic flows descending from Mayon Volcano.

Indigenous Knowledge

When asked about their indigenous knowledge relating to Mayon Volcano's eruptions, several villagers stated: Dakulon an palatandaan na aram mi tungkol sa pagtuga kang Mayon. – There are lots of warning signs related to Mayon's eruption." They mention that when rivers and creeks run dry, it is often an important forewarning of when Mayon will be scoffing down its dreaded lava. "Pag ubas an tubig na talagang diretso sa pirang bulan, aram mi na ma tuga na an Mayon. Tapos an lava an pighahaditan ming maray. – If the water runs dry for several straight months, we know that Mayon will be erupting soon. And lava forming from the volcano is the one we really worry about."

The villagers also talk about sparks coming from the volcano which rapidly create many sparkling rills among the crevices and signal the eruption of Mayon.

In addition, local farmers can hear roars and sense earthquakes unfelt by residents far from the volcano. As Domingo Arias, a barangay policeman, stated, "Pag may naguusok na kalayo hali sa Mayon, yan an señales na matuga na. Minsan nahahadit kami, minsan dai man. Sigeng tanog kang bulkan asin nakakadangog inot su mga para tanom ky maluyang daguldol. – If Mayon is spitting fiery sparks, that is the signal that it will erupt. Sometimes we worry about it, sometimes not. The volcano always makes low rumbling sounds and the farmers are the ones who hear it first." Totems are used to see if winds carry ashes.

Furthermore, animals such as baboy-damo (wild boars) and chickens live out their bionic senses too. Villagers have seen these animals scurry away from the volcano. When animals race recklessly from Mayon, it sends out

an alarm that it is time to evacuate because these animals can feel high temperature from the volcano.

According to its residents, native premonitions and superstitions regarding the eruption of Mayon Volcano have always been right. Queried about whether they worry about the approaching doom from the volcano when they see and feel these warning signs, they answer that they have gotten used to the situation. “Tuod na kami pagnagtutuga an Mayon, - We are immune to Mayon’s upsurge. “ coconut farmer Romeo Nantes, a father of three, commented. “We would not evacuate now unless it’s really urgent and there’s a big eruption,” Rosario Nantes, Romeo’s wife said, while tending to her little store. This is why in spite of mandatory evacuations, many of the villagers stayed put on their farms around Mayon to tend to crops and livestock while guarding homes and possessions.

A long-time resident of Matanag, Geronimo Toledo remarked, “Pag mauran, baha ang mas delikado pag nagtuga an Mayon. Pero pag maray an oras wara man dapat haditan. – If it rains, floods are more dangerous if Mayon erupts. But if the weather is fine, there is nothing to worry about.”

Such knowledge about the signs of the volcano’s spitting plumes of ash contributes to disaster risk reduction. By following these beliefs, locals have gathered warning signals from the impending threat and can be readily prepared. These locals are generally grateful that their elders have passed this knowledge to them.

“An mga gurang mi an nagturo samuya kang gabos na dapat ming maaraman. Maski aki mi aram an mga señales. – Our elders were the ones who taught us what we really ought to know. Even our children know the signals.” Arias affirmed.

When asked if they still pay attention to scientific announcements, Belga, another resident replies, “Dai kami nagtutubod sa awtoridad ta sala sinda minsan. Masabi na matuga pero wara man kaming napapansin na palatandaan o babala kaya dai kami mina hiro nangad hanggang sigurado kami. Pero pag aram ming tama, ma hali man sana kami siyempre. – We don’t really take note of what the authorities say sometimes. They tell us that Mayon will erupt but we don’t see any signals so we don’t really pull out unless we are really sure. But if we know that it’s true, we evacuate of course. “

“Mga para tanom sana kadkalan samo digdi pero aram mi kung ano an dapat gibuhan and tubodan. Sa pagtuga kang Mayon, sadiri ming kahiruan an kaipuhan. Aram mi yan, kaito pa. – Most of us are only farmers but we know what we should do and what to believe. Regarding Mayon’s eruption, we only rely on our own impulses. We have known that ever since.” Belga said almost stubbornly. With their indigenous knowledge, the villagers of Matanag do know to move out of harm’s way in case a major eruption occurs.

Lessons Learned

Officials often have trouble persuading villagers to clear out of danger zones despite natural vulnerabilities and disasters. Instead of listening to officials, geologists and volcanologists, people rely on spiritual advisers and credulous beliefs for guidance. People’s failure to evacuate in the face of fury-spouting, lava guzzling volcanoes just because of stubborn standpoints sometimes equates to absurdity.

Nonetheless, not all cultures view volcanic eruptions as destructive. Instead, many of those living near Mayon view its eruptions as beneficial acts of creation and evolution. Being near a volcano does not mean people cannot thrive on its surroundings. The local people know that volcanoes do not disgorge things out without reason. Volcanoes are often thought of as significant entities that wreak vengeance and impartiality to the world, bringing justice to life’s misgivings and wrongdoings of its inhabitants.

The differing views on volcanic eruptions produce a disconnection between scientists and the people who are directly affected. These views confound the scientific community but they cannot be dismissed immediately. In truth, the indigenous knowledge that has been passed from generation to generation helps them to

minimize risk, cope with natural dangers and learn how to survive. The level of risk people are willing to take because of their indigenous knowledge may be disbelieving. But these irrational tales and rituals also help these people cope with disaster. Social scientists observe that these are no longer naïve views of nature. Superstitions and myths are still grounded in people's core beliefs. These beliefs make them hopeful and unrelenting in the face of danger.

Scientists who use the latest technology can bridge the gap between the people who stand to benefit from their work. However, they must not underestimate the indigenous capabilities that the people possess and should understand how their work will be received by people who intimately experience and cope with volcanic eruptions.

Western Province, Solomon Islands

Indigenous Knowledge Saved Lives during 2007 Solomon Islands Tsunami

Brian G. McAdoo, Jennifer Baumwoll and Andrew Moore

Abstract

On 2 April 2007, an earthquake with a magnitude of 8.1 and its subsequent tsunami hit the Solomon Islands, killing 52 people. That number would have likely been considerably higher was it not for the appropriate reaction of the indigenous coastal populations. Immigrant populations died at disproportionate rates in comparably damaged areas because they did not recognize the signs of the impending tsunami. Both the indigenous and immigrant populations had little time to respond because their villages were very close to the earthquake's epicenter. While buoy-based early warning systems are necessary to mitigate the effects of trans-oceanic tsunamis that hit coastlines hours after an earthquake, they are limited in the near field regions close to the epicenter, especially those with developing infrastructure. Tsunami mitigation plans that seek to educate local populations must be cognizant of not only the physical environment of the region, but also economically and culturally diverse populations and their knowledge of the local environment.

Background

Between 1955 and the early 1960s, the ethnic demographics of the Western Province changed when a group of ethnic Gilbertese emigrated from their native Kiribati, a country of remote, low-lying coral atolls located in the southwest Pacific Ocean, far from any active earthquake sources. Due to resource depletion and population increase there, they came to the region as part of a British colonial resettlement scheme.¹ According to a 2002 Solomon Island government census, the Gilbertese constitute less than 4% of the country's population (~4000 of 495,000). The Gilbertese are traditionally atoll-dwellers that rely heavily on ocean resources, thus the majority of their settlements are located adjacent to the lagoons. Since the Gilbertese moved to the region there have not been any tsunamigenic earthquakes.²

¹ Matthew (1996).

² Fritz and Kalligeris (2008) report eyewitness accounts of small tsunamis on Ranongga generated by a local Ms=7.2 earthquake in 1959, and in Honiara on Guadalcanal in 1952 that likely resulted from a distant earthquake in Kamchatka in the far NW Pacific.

Story/Event

On 2 April 2007 at 7:39AM local time, a magnitude 8.1 earthquake hit the Solomon Islands' Western Province.³ The earthquake caused strong shaking (hard enough that most people had problems remaining standing) which lasted over 1 minute, damaging structures and shearing off coral reef colonies,⁴ while at the same time warning the populace of the potential for a tsunami. Structures in the region were significantly damaged, and injuries resulted from falling building materials and burning water from cook stoves.⁵ The earthquake also triggered over 1,000 landslides on the steep volcanic island of Ranongga- one occurred at the village on Mondo, killing 2 people.⁶ Strong shaking and coseismic uplift also damaged the delicate coral reef in the lagoons, which will have lasting effects on the recovery of the fisheries on which both the indigenous Melanesian and immigrant Gilbertese populations rely.

³ USGS (2007).

⁴ McAdoo et al. (2008).

⁵ Fritz and Kalligeris, (2008).

⁶ McAdoo et. al. (2008).

Almost immediately after the shaking stopped, according to eyewitnesses, water rushed out of the shallow coral lagoon, leaving the seafloor exposed. The tsunami came back in between 3 and 10 minutes after the shaking ceased. Anecdotal evidence, supported by geologic observations, describes two or three subsequent waves that came in as relatively low power, rapidly-rising tides rather than fast-moving turbulent bores.⁷ Low density coral colonies with large surface areas that the earthquake sheared off at their bases landed near their original growth positions, and would have been moved in the presence of a strong current. The tsunami picked up vehicles and houses, floating them inland and dropping them with very little damage—more powerful bores tend to roll vehicles and dismantle structures. The tsunami was responsible for 50 of the 52 deaths caused by this disaster. Figure 1 illustrates the geographic distribution of runup heights as they relate spatially to deaths.

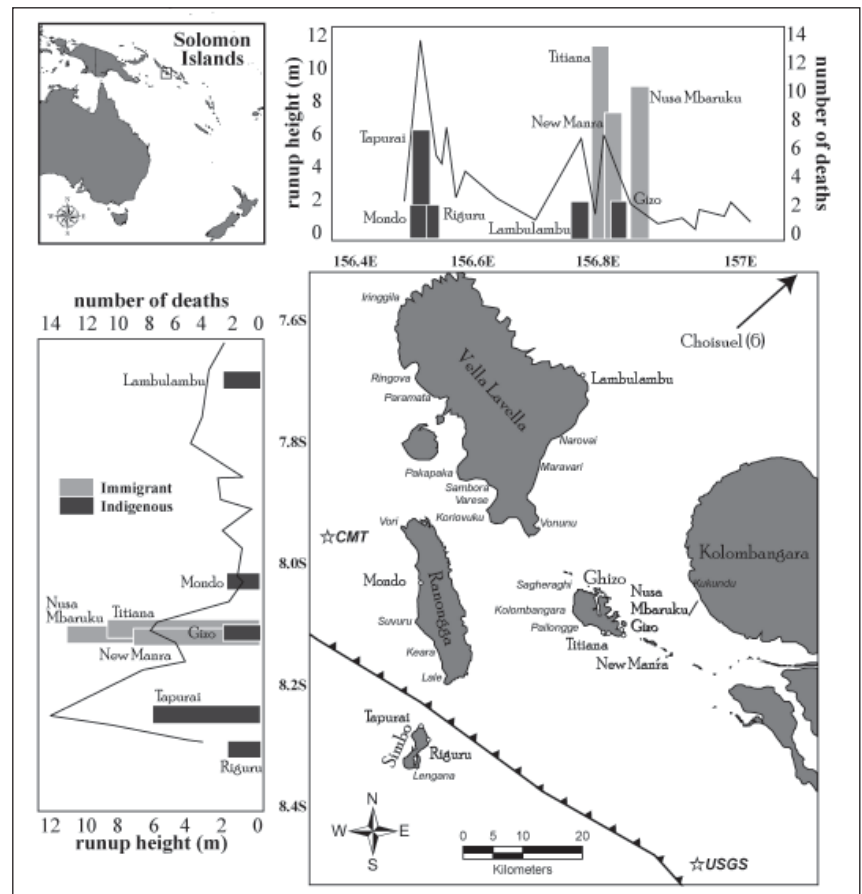


Figure 1. Geographic distribution of runup heights from the 2 April 2007 Solomon Islands tsunami as they relate spatially to deaths. Villages on the map are those that suffered significant damage, and those in larger bold type recorded deaths. There tend to be more deaths where the tsunami was larger, however most of those deaths occurred in the minority, immigrant populations. (After McAdoo et al., in press, Natural Hazards)

⁷ McAdoo et al. (2008).

Indigenous Knowledge

The geologic survey found that a tsunami that came ashore at the same height in similar physical surroundings in areas with different demographics produced inconsistent mortality patterns which could not be explained by the physical hazards alone—Gilbertese immigrants died at disproportionate rates compared to the indigenous Melanesians.⁸ While the immigrant villages tend to have higher populations, their response to the earthquake, which was to investigate the emptied lagoon, demonstrated a lack of understanding of how tsunamis behave. Each of the villages had a good coral reef/lagoon buffer, and one of the affected immigrant villages (New Manra) even had an additional stand of mangroves that the indigenous communities

lacked. This evidence suggests that the primary factor contributing to increased mortality was a proper response based on generations accustomed to living on an active island arc adjacent to a subduction zone.

The immigrant community of Titiana is only 3 km east of Pailongge on the south coast of Ghizo Island that took the brunt of the tsunami. The broad lagoon (100–400 m wide) lies behind a steep coral reef front, like Pailongge, and reflected some of the wave energy back to sea.⁹ According to survivors, however, when the shaking stopped and the ocean receded from the lagoon, curious children from the village went down to explore the exposed seafloor. In the Gilbertese villages of New

⁸ McAdoo et al. (in press).

⁹ McAdoo et al. (2008).

There were a lot of heroic stories that emerged after the disaster; stories of women and men who dashed back into the raging sea to save their love ones. Some succeeded, most however lost their lives in their final act to save lives.

*Gizo resident
Robertson Szetu
(Galathea Expedition)*

Manra, Titiana and Nusa Mbaruku, 67.7% of those that died were children. In Titiana alone, 8 of the 13 people that died were children who were not strong enough to swim against the incoming waves. As described in Figure 2, the immigrant villages recorded more deaths in Ghizo Island.

Anecdotal evidence suggests that many adults here were overwhelmed by the tsunami as they went to rescue their children-

On the other hand, there were no deaths in the indigenous Solomon Island village of Pailongge (population 76) on Ghizo's south coast. The sea withdrew almost immediately after the earthquake shaking ceased, according to eyewitnesses. The geologic evidence suggests that the steep reef front and broad lagoon (100-500 m wide) slowed the incoming tsunami,¹⁰ however the wave was still quite large when it came ashore, flowing overland around 3 m deep.¹¹ After the shaking stopped, village elders noticed that the lagoon had emptied, helped direct everyone inland, and heads of households made sure that children were accounted for and evacuated.

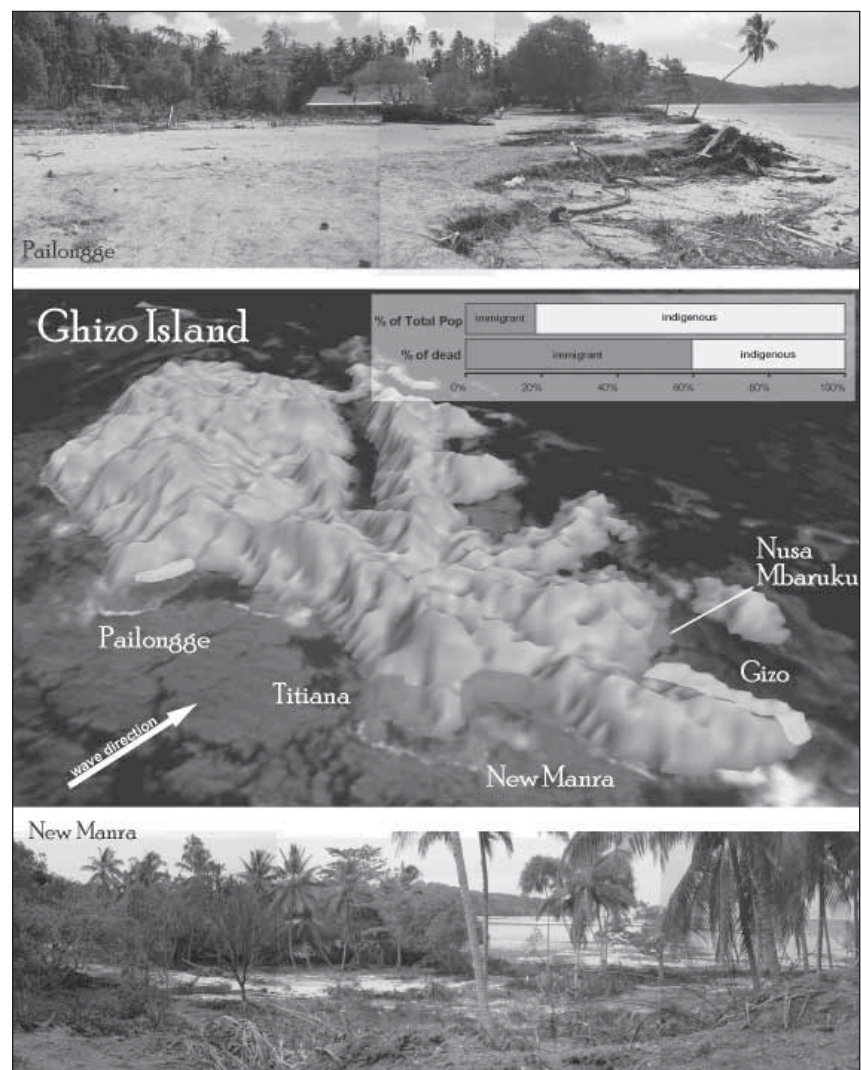


Figure 2. Gizo has a population of indigenous Solomon Islanders (yellow) and immigrant Gilbertese (red) that live in largely segregated villages. The indigenous village of Pailongge lies on the south coast exposed to the brunt of the tsunami, as was the immigrant village of Titiana, and both villages have accessible high ground for escape. No one died in Pailongge, and only 2 died in the more populous Gizo, where each of the immigrant communities recorded deaths regardless of geography. New Manra, an immigrant community, had not only a protective reef, but also a stand of mangroves in front of the village, yet 8 people (of 206) died. (after McAdoo et al., submitted to Natural Hazards)

¹⁰ McAdoo et al. (2008).

¹¹ Fritz and Kalligeris (2008).

Lessons Learned

Of the 52 people that died during the Solomon Islands earthquake and tsunami, 31 (59.6%) were immigrant Gilbertese from Titiana, New Manra and Nusa Mbaruku that did not react properly because they had no memory in their culture of such an event.¹² Kiribati is a coral atoll nation, situated far from any regular earthquake sources. Because there have been no major, tsunamigenic earthquakes in the 50 years since their emigration, they simply lacked the indigenous knowledge of their adopted environment that could have helped save their lives. Gilbertese children were particularly vulnerable because not only were they too weak to swim against the relatively slow-moving yet deep tsunami, but they lacked the indigenous knowledge that would have kept them from exploring the emptied lagoons. The indigenous Solomon Islanders, on the other hand, in large part responded in a way that reduced their overall mortality.

Indigenous knowledge of the Solomon Islands, where active volcanoes and earthquakes are more common, mitigated the effects of this tsunami. In indigenous villages on hard-hit Ghizo Island, the effects of the tsunami were mitigated by the combination of 1) a healthy coral reef with a steep barrier front and wide, shallow lagoon that reflected and attenuated some of the tsunami's energy, 2) accessible and effective escape routes and high ground provided by the existing topography, and 3) an indigenous knowledge of what to do during a strong earthquake followed by an emptying lagoon.¹³ Immigrant Gilbertese villages with both the same physiography and who were hit by a tsunami of equal intensity, lacked indigenous knowledge which led them to suffer more casualties. Many people died at the indigenous village on Tapurai, which lacked an effective coral reef barrier due to its natural morphology that evolved on the lee side of Simbo Island.

Indigenous knowledge is an effective tsunami mitigation tool when the right combination of education and physiography come together. Locations with broad coastal plains would have a hard time evacuating the coast, especially if population densities are high as is the case with Banda Aceh, Indonesia during the 2004 Indian Ocean tsunami. Nonetheless, a barrier reef, wide lagoon and stand of mangroves were not enough to protect the residents of New Manra since they had no knowledge of tsunamis in this region.

¹² McAdoo et al. (in press).

¹³ McAdoo et al. (in press).

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Surin Islands, Thailand

Saved by an Old Legend and a Keen Observation: The Case of Moken Sea Nomads in Thailand

Narumon Arunotai

Abstract

There is a lesson to be learnt in the case of the Moken sea nomads in Thailand – that the Moken have indigenous knowledge that enable them to reduce risk from disaster. It could be said that the Moken were saved by an old legend, a keen observation of the sea and a strong sense of precaution, their intimate knowledge of marine and forest environment, their clever selection of village site, and their boat maneuvering skill. This article explores lessons from Moken indigenous knowledge in risk reduction and the implications for applying these lessons to relevant policy and practices.

“We call this big wave ‘laboon’. It is the cleansing wave that came to wash the dirty shore. It is the god of wave, a furious one that consumes and destroys. I had never seen such wave, only heard it from the ancient ones. They talked about the ‘seven roller waves’ that visits us once in every two generations.”

Salama Klatalay, Moken elder

Background

The Moken, the “sea nomads” or “sea gypsies” of the Andaman Sea, are a former marine nomadic group generally known in Thailand as Chao Lay (sea people). During the dry north-east monsoon season when the sea is calm the Moken lived in traditional boats called kabang, traveling to islands in the Mergui Archipelago in the Andaman Sea to make a living. The social structure consists of kin groups of about two to ten families traveling together. In the southwest monsoon season the sea is rough and unpredictable, so the Moken adapt to this by living in temporary huts in protected bays and making their livelihood close to their villages.

The large Moken communities in Thailand can be found in the islands of Lao, Sinhai, Phayam, and Chang in Ranong Province, the Surin Islands in Phang-nga Province, and in Rawai Beach in Phuket Province. The Moken from the Surin Islands remain relatively traditional compared to the other groups. Even though they have started living in the same place for a longer period, they still use forest materials for their huts

and occasionally move their village to another site. Nowadays there are about 2,000 Moken in the Mergui Archipelago in Myanmar and about 800 Moken in Thailand.¹

The Surin Islands have been the home and the foraging ground for the Moken for centuries. The islands are located in the Andaman Sea, approximately 60 kilometers off the southwest coast of Thailand. Administratively, the Surin Islands are part of Khuraburi District in Phang-nga Province, which lies approximately 720 kilometers southwest of Bangkok.² Phang-nga is one among six provinces that received the most devastated impact from the tsunami.

The Surin Islands' main natural features are coral reefs and forests. The coral reef surrounding the Surin Islands is reported to be the largest and widest in Thailand. Tropical rainforest is the other major natural feature on the islands. It accounts for over 90 percent of the forested area. The forested areas are well preserved and play an important role in the Moken's traditional lifestyle.³ The Surin Islands were declared Thailand's twenty-ninth national marine park in 1981. Facilities for tourists have been gradually developed and during 2003 the Park received over 36,000 tourists during their 6 months opening season (from mid November to mid May).

Prior to the tsunami, there were 2 Moken communities on the Surin Islands. One community, consisting of 16 households, was located at Sai-En Bay on North Surin Island. The other community, consisting of 30 households, was located at Small Bon Bay on South Surin Island. The Moken population on the Surin Islands fluctuated seasonally and annually. Some individuals and families migrated between the Surin Islands and other islands in Myanmar waters and between the two communities within the Surin Islands. During 2004, there were 184 Moken living on the Surin Islands. Of these, 77 were male and 107 were female. About half of the population is 18 years old or younger, and children under the age of ten constitute about one-third of the population.⁴ As of February 2008, there are about 220 Moken on South Surin Island, living in one large village at Large Bon Bay.

Story/Event

On the morning of Sunday December 26, 2004, the Moken of the Surin Islands observed a sudden change in the sea level. This occurred without any change of weather and it was considered a very unusual phenomenon. For several Moken elderly, it signaled the coming of "seven rollers", a legend that has passed down for generations. The whole community ran up to the hill behind the village very quickly, and all survived the tsunami disaster, though the entire village was swept away along with a few boats.

At the same time, over 20 Moken had been working at the National Park. Those who took tourists out for snorkeling noticed a change in the current and decided to steer the boats away from the shore. Due to their sharp instinct and outstanding boat maneuvering skill, they saved the lives of many tourists. Again, it was the Moken who took the National Park staff and tourists up their foraging trails to seek a safe place to spend the night while waiting for a bigger boat to take them all to shore.

After their two villages were swept away by the tsunami, the Moken of the Surin Islands came to shore to take refuge in the local temple. Within two weeks, they missed the islands and the sea, and felt confident enough to move back to the Surin Islands. Since they are a group that accumulates few material possessions, they did not dwell on the past, but continue with their daily lives as in pre-tsunami days.

⁴ Arunotai, Wongbusarakum, and Elias (2007): 12.



Figure 1. The Moken and their kabang. Photo: Paladej Na Pombejra

¹ Arunotai (2006):140

² Arunotai, Wongbusarakum, and Elias (2007):8-9.

³ Arunotai, Wongbusarakum, and Elias (2007):8-9

Indigenous Knowledge

The Moken of the Surin Islands have become widely known in Thailand and internationally as the group of people who escaped from the tsunami unscathed. The Moken survived not only because of the old legend of the seven rollers, but also because of their keen sense of observation, their alertness and sense of precaution, their intimate knowledge of marine and forest environment, the clever selection of their village site, and their boat maneuvering skill. The details of their knowledge, keen sense, and relevant skills are as follows:

1. Legend of the seven rollers. The Moken are a non-literate group but their “oral literature” is creative and rich. Living on islands and coastal areas for centuries, their ancestors must have experienced tsunami incidents, and they cleverly encoded the danger of tsunami disaster in a legend of the laboon or giant wave. The legend teaches that the laboon which usually comes as series of waves, hence “seven rollers”. While openly talking about laboon was forbidden out of fear that it would bring the deadly wave upon them, everyone knew that when the water along the shore suddenly went dry, they had to run to higher ground to save their lives. This knowledge represents the most effective natural disaster warning system, without any help from modern technology.⁵ The tsunami warning sign is imprinted in their cognitive system, so they are all able to survive even though most have not even seen a tsunami before.
2. Keen sense of observation and precaution. The Moken are very observant, especially regarding the change in weather and other unusual occurrences. They use the principle of precaution and are always alert. This is due to the risk encountered in everyday lives – from natural risks like unexpected storm, dangerous sea and forest animals, to human-related risks which can be traced from the Moken’s long history of being abused and exploited by outsiders – pirates, slave-raiders, robbers, bandits, etc.⁶ Therefore, as soon as sea water receded in an unusual way on December 26, 2004, the Moken flagged one another to seek refuge on the hill behind the village, and those who were taking tourists out for snorkeling headed their boats away from the shore to avoid impact.
3. Intimate knowledge of marine environment. Being the “sea people” who live off, from, and by the sea, the Moken have a deep-seated knowledge of the natural rhythms, an ability to recognize natural signs and to distinguish between “usual” and “unusual” phenomena. For them, the mass of water receding very quickly was peculiar, and the incoming white breaker was definitely a sign of laboon as it occurred without any stirring wind or a changing sign in the sky.

⁵ Arunotai (2006):143

⁶ Hinshiranan (1996):131-133



Figure 2. Left: Tsunami sign seen throughout the southern coast of Thailand. Right: Such a sign can be “imprinted” in the young through indigenous knowledge.

4. Knowledge of forest trails and survival in the wild. After running uphill to seek refuge from the tsunami impact, the Moken know “where” to run to because they are familiar with the forest environment and terrain.
5. Clever selection of village site. The Moken who reside on different islands within Thailand’s Andaman Sea carefully select an appropriate site for their villages, that is, the area on the eastern part of the islands. A comparative analysis of indigenous settlements (Moken’s and Urak Lawoi’s) leads to the conclusion that each of the settlements are in a protected bay on the east. This is because the islands in the Andaman Sea are influenced by two monsoons, southwestern monsoon which brings rain, strong winds, waves, and storms; and northeastern monsoon which brings drier weather and milder winds. Having settlements on the eastern side of the island means being relatively well-protected from southwestern wind (and thus from tsunami from the high sea).⁷ For the Moken of Surin Islands, the high mountain behind the village also provides enough high ground for evacuation.

Moken traditional settlement, village, and huts, including beliefs and practices about hut construction, all reflect traditional knowledge which enables the Moken to reside comfortably and safely in the coastal environment. In addition, a small village with long-stilted huts situated on the water has been a significant part of Moken cultural identity.⁸ It should be noted here that their tsunami survival is partly due to “keen observation” – from traditional village layout, most Moken are able to observe the sea right from their hut.

6. Boat maneuvering and other marine-related skills. Moken men have very good boat maneuvering skills because they become familiar with rowing smaller boats and operating the motor for larger boats at an early age. In turbulent waters, the men can skillfully steer boats away from shore. Both men and women also have other marine-related skills such as swimming and diving. Furthermore, men and women, young and old, relatively maintain their physical fitness as their work usually involves physical strength. Even the elderly and children are able to run and climb up to higher ground without much difficulty.

With regard to the passing on of indigenous knowledge, there is no means of documenting or transforming knowledge into the textual form that people can read or follow because the Moken language has only verbal form, no script or written language. From a modern education perspective, this poses a limitation as there is no means to record the knowledge and no other “short-cut” to that knowledge except by learning it directly from those who know through, 1) observation, 2) memorization or bearing in mind, 3) trial or hands-on, 4) discovery-exploring-reinventing, and 5) practicing and reviewing.

In the eyes of the sedentarized land-based population of Thailand who have permanent homes, the nomadic and simple lives of the Moken seem to be outdated, backward, and without potential for progress.⁹ This is likely to be a social representation that the Moken themselves have adopted from mainstream society.¹⁰ As a result, the Moken indigenous knowledge is rarely recognized or valued.

⁷ Arunotai and Elias (2005).

⁸ Arunotai and Elias (2005).

⁹ Arunotai (2003):116–117.

¹⁰ Arunotai (2006):141.



Figure 3. Pre-tsunami huts, with entrance facing inland but with back porch overlooking the sea.

Lessons Learned

The major lesson learnt here is that traditional Moken knowledge previously regarded as “common” or even “outdated” enabled the Moken (and others) to survive the tsunami disaster. It is unfortunate that these forms of knowledge are now limited only to the adults and the elderly. They are gradually forgotten and rarely passed on to the young generations.

As for the legend and folktales, “Some of the tales are merely short versions of longer tales, and oral literature no longer has much meaning for them.”¹¹ “The traditional indigenous knowledge of the Moken is not recognised as knowledge or science; thus the direction of change is towards the kind of development defined and induced by outsiders who see them as backward and poor.”¹²

There are several implications for applying these lessons to relevant policy and practices.

1. To value and promote indigenous knowledge through the development of relevant local curriculum and community participation in education management. Children should have a chance to learn from the elderly in the community, to absorb and practice “keen observation” and the “precautionary principle”, and to learn and cherish intimate knowledge of marine and forest environment. Formal education places an emphasis on “explicit” knowledge, but such “tacit” knowledge and skills have proved to be crucial to the Moken’s physical and cultural survival.
2. To “internalize” early warning signs and disaster preparedness plans like evacuation, escape routes, and getting to emergency shelter. For the Moken, the coming of laboon is actually imprinted in their cognitive system although it was previously half-myth and half-reality. Not only were they able to recognize warning signs, they were also familiar with the forest trails which they used as their escape routes and staying/surviving in the wild was not a problem for them either. For other communities, apart from having educational materials on natural disaster for communities and schools, there needs to be regular practices and drills for different kinds of possible disaster or occurrence in the area.
3. To learn important survival knowledge and skills like swimming, climbing, first-aid, etc. One should keep physically fit in order to be prepared. The Moken community is small, so they know one another well and know who needs special help. For other communities, there should be a mapping of the elderly, people with disabilities, and others who need special help so that they can be taken care of during the emergencies.

¹¹ Ivanoff (2001):3.

¹² Arunotai (2006):148

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